

# Share of Wallet Intelligence Engine — Syn Bank

Team **The Independent Variable** · Daniel Mataranyika · Herton Mabongue · Standard Bank × Data School Hackathon 2026

4.5%  
SYN BANK SHARE

R4.29tn  
ADDRESSABLE FLOW

R4.10tn  
UNCAPTURED GAP

R9.65bn  
FEE OPPORTUNITY / YR

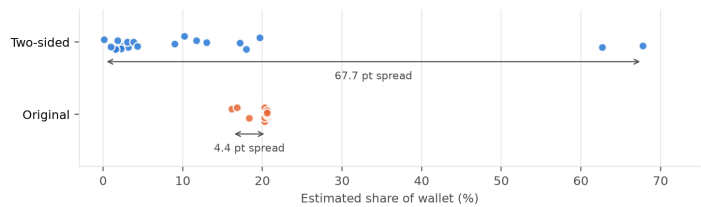
## THE PROBLEM, AND THE TRAP IN IT

Syn Bank sees every transaction that flows through Syn Bank — and none of the ones that flow through its competitors. Share of wallet therefore cannot be read off the internal data; it must be estimated from outside.

Our first model missed this. It sized the wallet from internal flow, then took captured volume as a fraction of that wallet:

$$\text{share} = (\text{wallet} \times \text{capture\_ratio}) / \text{wallet} = \text{capture\_ratio}$$

The wallet term cancels. Share was identical to the assumed capture rate and could not depend on the data. It passed every obvious check — values between 0 and 1, a plausible ranking — while giving **20 very different companies the same answer**: 4.4 points of spread across a portfolio whose wallets differ by 2,757×.



## THE TWO-SIDED MODEL

The two sides of the ratio are now computed from independent sources.

$$\text{share} = \text{captured flow (measured)} / \text{addressable flow (estimated)}$$

- Top-down** — addressable flow sized from each client's published FY2025 statements: revenue drives transactional, foreign revenue drives FX, cost of sales plus inventory drives trade finance. Each scaled by an explicit SA-attribution share, since Syn Bank is a South African bank.
- Bottom-up** — captured flow summed over the trailing 12 months of the internal datasets. No assumption enters this side.

Share now spans **0.1% to 67.8%** (67.7 points). Every external figure is cited per client in data/external/company\_financials.csv.

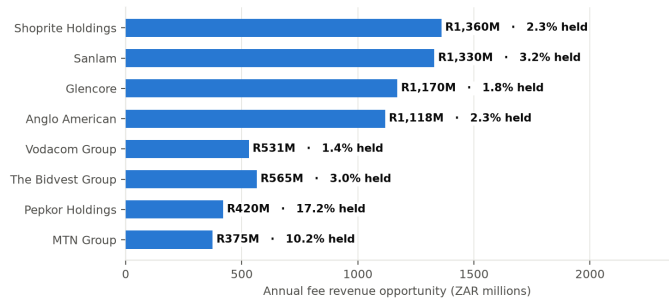
## GENERATIVE AI, DOING REAL WORK

The model never supplies a number. Figures are computed by the engine, passed in as an evidence block, and reused verbatim — the model does language, not arithmetic. Every output is then re-read by a grounding check that flags any rand amount or percentage absent from the evidence. It writes pre-meeting briefings and answers open questions over the portfolio, and degrades to deterministic text if the API is unavailable, so a demo never breaks. Prompts and outputs: prompts/logs/.

## WHAT A COVERAGE BANKER DOES WITH IT

| # | CLIENT            | SECTOR         | SHARE | FEE OPPTY/YR | LEAD WITH     |
|---|-------------------|----------------|-------|--------------|---------------|
| 1 | Shoprite Holdings | Consumer Goods | 2.3%  | R1,360M      | Transactional |
| 2 | Sanlam            | Insurance      | 3.2%  | R1,330M      | Transactional |
| 3 | Glencore          | Mining         | 1.8%  | R1,170M      | Transactional |
| 4 | Anglo American    | Mining         | 2.3%  | R1,118M      | Transactional |
| 5 | Vodacom Group     | Telecoms       | 1.4%  | R531M        | Transactional |

Ranking blends gap size (50%) with ease of conversion, relationship depth and 90-day urgency, so the list is ordered by winnable value rather than client size. **Transactional banking is the lead pillar for 19 of 20 clients** — cash-management consolidation is the repeatable conversation.



## HOW MUCH DO THE ASSUMPTIONS MATTER?

Shifting every flow multiple by ±25% moves the **portfolio-wide average** share only between **3.6% and 5.9%** (not the 0.1%–67.8% per-client range above); the captured side cannot move, because it is measured. Our multiples imply banking fees of ~0.47% of revenue, inside the 0.4–0.7% range the brief itself cites.

## LIMITATIONS

20 clients supplied, not the 50 described. Flow multiples are benchmarks, not observations, and SA-attribution is the most judgmental input. Revenue basis differs by industry and is recorded per client; cost of sales and inventory are imputed from sector margins where not disclosed, and flagged. Gaps are attributed to competitors, assuming internal records are complete. Lending and DCM are out of scope.

Public financial statements are used only to size an addressable benchmark. No synthetic record is linked to a real entity or event, and Syn Bank is treated throughout as fictional. Full method and citations: docs/methodology.md.